

2005

**HIGHER SCHOOL CERTIFICATE
EXAMINATION**

Chemistry

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen
- Draw diagrams using pencil
- Board-approved calculators may be used
- A data sheet and a Periodic Table are provided at the back of this paper
- Write your Centre Number and Student Number at the top of pages 13, 15, 17, and 21

Total marks – 100

Section I Pages 2–24

75 marks

This section has two parts, Part A and Part B

Part A – 15 marks

- Attempt Questions 1–15
- Allow about 30 minutes for this part

Part B – 60 marks

- Attempt Questions 16–27
- Allow about 1 hour and 45 minutes for this part

Section II Pages 25–36

25 marks

- Attempt ONE question from Questions 28–32
- Allow about 45 minutes for this section

Section I

75 marks

Part A – 15 marks

Attempt Questions 1–15

Allow about 30 minutes for this part

Use the multiple-choice answer sheet.

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample: $2 + 4 =$ (A) 2 (B) 6 (C) 8 (D) 9
 A ☐ B ☒ C ☐ D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

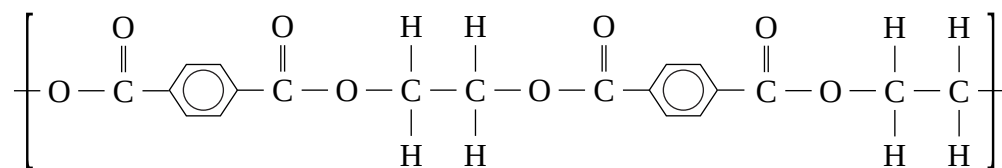
A ☒ B ☒ C ☐ D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate the correct answer by writing the word *correct* and drawing an arrow as follows.

A ☒ B ☒ C ☐ D ☐
 correct
 ↙

- 1 Which of the following conditions would produce a radioactive isotope?
- (A) Too many atoms in the sample provided
 - (B) Too many protons and neutrons in the atom
 - (C) Too many electrons in the outer shell of the atom
 - (D) Too many electrons for the number of neutrons in the atom
- 2 Which of the following makes cellulose a potential raw material for the petrochemical industry?
- (A) It is available in many parts of the world.
 - (B) It is light in weight and easy to transport.
 - (C) It is an organic polymer based on carbon.
 - (D) It is a polymer that undergoes combustion.
- 3 The heat of combustion of butan-1-ol is 2676 kJ mol^{-1} .
What is the value of the heat of combustion in kJ g^{-1} ?
- (A) 30.41
 - (B) 36.10
 - (C) 44.60
 - (D) 47.79

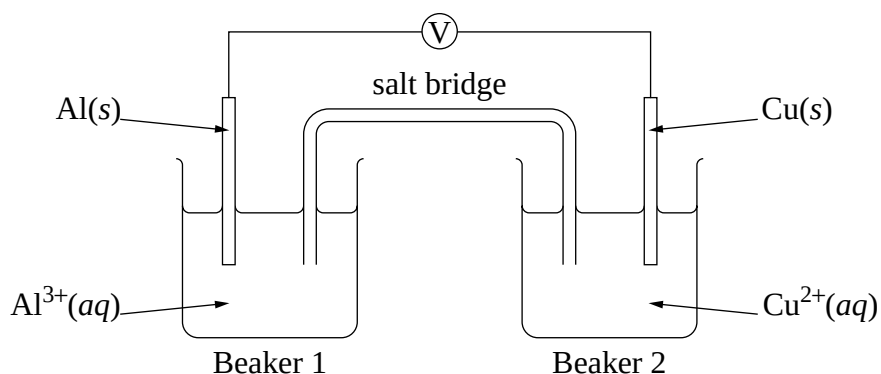
- 4 Terylene (polyester) is a condensation polymer. Part of the structure of the polymer is shown.



What are the two monomers that form this polymer?

	<i>Monomer 1</i>	<i>Monomer 2</i>
(A)	$\text{H}-\text{C}_6\text{H}_4-\text{H}$	$\text{HO}-\underset{\underset{\text{H}}{\mid}}{\overset{\text{H}}{\mid}}\text{C}-\underset{\underset{\text{H}}{\mid}}{\overset{\text{H}}{\mid}}\text{C}-\text{OH}$
(B)	$\text{HO}-\underset{\underset{\text{H}}{\mid}}{\overset{\text{H}}{\mid}}\text{C}=\text{C}-\text{OH}$	$\text{HO}-\overset{\overset{\text{O}}{\parallel}}{\text{C}}-\text{C}_6\text{H}_4-\overset{\overset{\text{O}}{\parallel}}{\text{C}}-\text{OH}$
(C)	$\text{H}-\text{O}-\overset{\overset{\text{O}}{\parallel}}{\text{C}}-\text{H}$	$\text{H}-\text{C}_6\text{H}_4-\text{H}$
(D)	$\text{HO}-\overset{\overset{\text{O}}{\parallel}}{\text{C}}-\text{C}_6\text{H}_4-\overset{\overset{\text{O}}{\parallel}}{\text{C}}-\text{OH}$	$\text{HO}-\underset{\underset{\text{H}}{\mid}}{\overset{\text{H}}{\mid}}\text{C}-\underset{\underset{\text{H}}{\mid}}{\overset{\text{H}}{\mid}}\text{C}-\text{OH}$

- 5 An electrochemical cell is set up as shown in the diagram.



What are two observations for this electrochemical cell?

	<i>Observation 1</i>	<i>Observation 2</i>
(A)	A reading was shown on the voltmeter	In Beaker 2 the solution became a darker blue
(B)	In Beaker 2 the blue solution faded	A reddish precipitate formed on the copper electrode
(C)	A grey precipitate formed on the aluminium electrode	In Beaker 2 the solution became a darker blue
(D)	Electrons moved through the voltmeter	A reddish precipitate formed on the copper electrode

- 6 Which of the following is a common use for ethyl pentanoate?

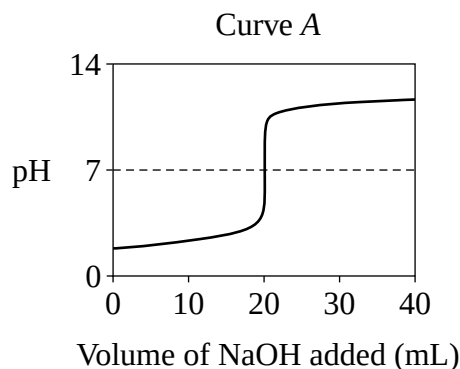
- (A) Flavouring
- (B) Fuel
- (C) Indicator
- (D) Solvent

- 7 A series of observations was made about a first-hand investigation.
- A cup of carbonated soft drink was poured from a newly-opened, chilled bottle.
 - A lot of bubbles were observed in the cup.
 - The cup was then left on the kitchen bench for several hours.
 - After this time, it was observed that there were no bubbles visible in the cup.

Why were there no bubbles visible in the cup after several hours?

- (A) The bubbles of carbon dioxide had dissolved into the soft drink.
 - (B) The carbon dioxide had been neutralised by the water in the soft drink.
 - (C) Carbon dioxide is less soluble at higher temperatures and lower pressures.
 - (D) Carbon dioxide is more soluble at higher temperatures and lower pressures.
- 8 What would be the pH of a 0.1 mol L^{-1} solution of sulfuric acid?
- (A) Less than 1.0
 - (B) Exactly 1.0
 - (C) Between 1.0 and 7.0
 - (D) Greater than 7.0
- 9 Which of the following pairs would form a buffer solution?
- (A) $\text{HCl}(aq) / \text{Cl}^-(aq)$
 - (B) $\text{H}_2\text{PO}_4^-(aq) / \text{PO}_4^{3-}(aq)$
 - (C) $\text{H}_2\text{SO}_4(aq) / \text{HSO}_4^-(aq)$
 - (D) $\text{CH}_3\text{COOH}(aq) / \text{CH}_3\text{COO}^-(aq)$

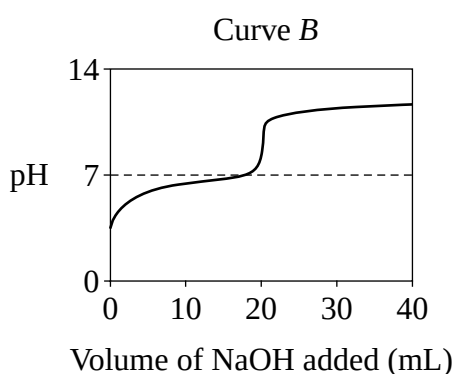
- 10 A titration was conducted by adding NaOH from a teflon-coated burette to HCl in a conical flask. The pH in the flask was recorded during the titration and Curve A was produced.



The table shows appropriate indicators used to identify the equivalence point in titrations. For NaOH and HCl the appropriate indicator is bromothymol blue.

<i>Indicator</i>	<i>Acidic colour</i>	<i>Range of colour change</i>	<i>Basic colour</i>
Methyl orange	red	3.1 – 4.4	yellow
Methyl red	red	4.4 – 6.2	yellow
Bromothymol blue	yellow	6.0 – 7.6	blue
Cresolphthalein	colourless	8.1 – 9.7	red
Alizarin yellow	yellow	10.1 – 12.0	red

A second titration was conducted by adding NaOH to a different acid. The pH in the flask was recorded during the titration and Curve B was produced.



What is the appropriate indicator for Curve B using the table?

- (A) Methyl orange
- (B) Methyl red
- (C) Cresolphthalein
- (D) Alizarin yellow

11 Which of the following substances is best analysed by atomic absorption spectroscopy (AAS)?

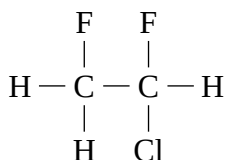
- (A) Calcium
- (B) Iodine
- (C) Nitrogen
- (D) Silicon

12 A student performed a sampling technique as part of a first-hand investigation on water quality and repeated the sampling technique several times.

What aspect of the experiment was improved by repeating the procedure?

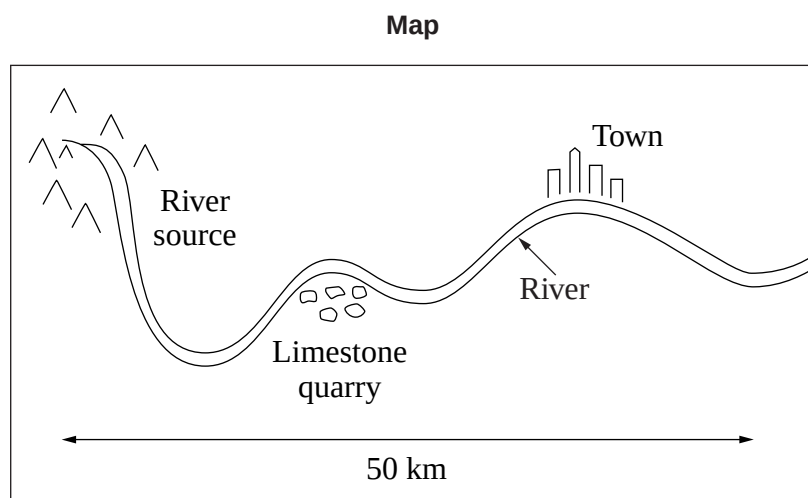
- (A) Accuracy
- (B) Reliability
- (C) Safety
- (D) Validity

13 What is the name of the following compound?



- (A) 1,2-difluoro-1-chloroethane
- (B) 1,2-difluoro-2-chloroethane
- (C) 1-chloro-1,2-difluoroethane
- (D) 2-chloro-1,2-difluoroethane

- 14 Water quality analyses were performed at different sites on a river shown on the map.



The table shows the results of the tests performed at the different sites.

Results of tests performed			
Site	Dissolved oxygen (ppm)	Biochemical oxygen demand (ppm)	pH
1	3	145	5.2
2	10	2	6.9
3	7	12	7.4

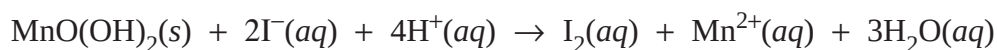
Where was each set of tests performed?

	River source	Limestone quarry	Town
(A)	Site 2	Site 1	Site 3
(B)	Site 1	Site 2	Site 3
(C)	Site 1	Site 3	Site 2
(D)	Site 2	Site 3	Site 1

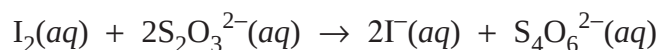
- 15 The Winkler method is used to determine the amount of dissolved oxygen in a sample. In this procedure, oxygen reacts with Mn^{2+} under alkaline conditions to produce a precipitate of $\text{MnO}(\text{OH})_2$.



The precipitate is then dissolved in acid and reacted with iodide, forming iodine and Mn^{2+} .



Finally, the amount of iodine produced is determined by reaction with thiosulfate.



When a sample of water was analysed using the Winkler method, a total of 0.60 mol of thiosulfate was used in the reaction.

How many moles of oxygen were present in the original sample?

- (A) 0.15
- (B) 0.30
- (C) 0.60
- (D) 1.20

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Centre Number

Section I (continued)

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Student Number

Part B – 60 marks

Attempt Questions 16–27

Allow about 1 hour and 45 minutes for this part

Answer the questions in the spaces provided.

Show all relevant working in questions involving calculations.

Marks

Question 16 (5 marks)

During your practical work you performed a first-hand investigation to distinguish between an alkene and the corresponding alkane.

- (a) Name the alkene used in your investigation. 1

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- (b) Identify a potential hazard in your investigation, and outline how you addressed this hazard. 2

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- (c) Outline the procedure you used for your first-hand investigation. 2

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Question 17 (3 marks)

The heat of combustion of ethanol is 1367 kJ mol^{-1} . In a first-hand investigation to determine the heat of combustion of ethanol, the experimental value determined differed from the theoretical value.

- (a) Identify a reason for this difference. 1

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- (b) Calculate the theoretical mass of ethanol required to heat 200 mL of water from 21.0°C to 45.0°C . 2

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Question 18 (4 marks)

Assess the impact of the recent development of a named biopolymer on society and the environment. 4

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Centre Number

Section I – Part B (continued)

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Student Number

Marks

Question 19 (4 marks)

You have studied one of the cells shown below.



Cell X



Cell Y

Choose ONE of the cells and answer parts (a) and (b).

- (a) State ONE environmental impact associated with the cell you have chosen. 1

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- (b) Describe the chemistry of the cell you have chosen. 3

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Describe the chemistry and procedure involved in each of these steps, using diagrams where appropriate.

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Centre Number

Section I – Part B (continued)

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Student Number

Marks**Question 21** (5 marks)

Analyse how knowledge of the composition and properties of acids has led to changes in the definition of acids.

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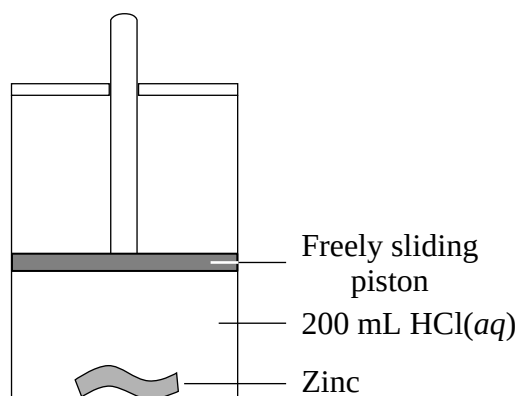
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Question 22 (7 marks)

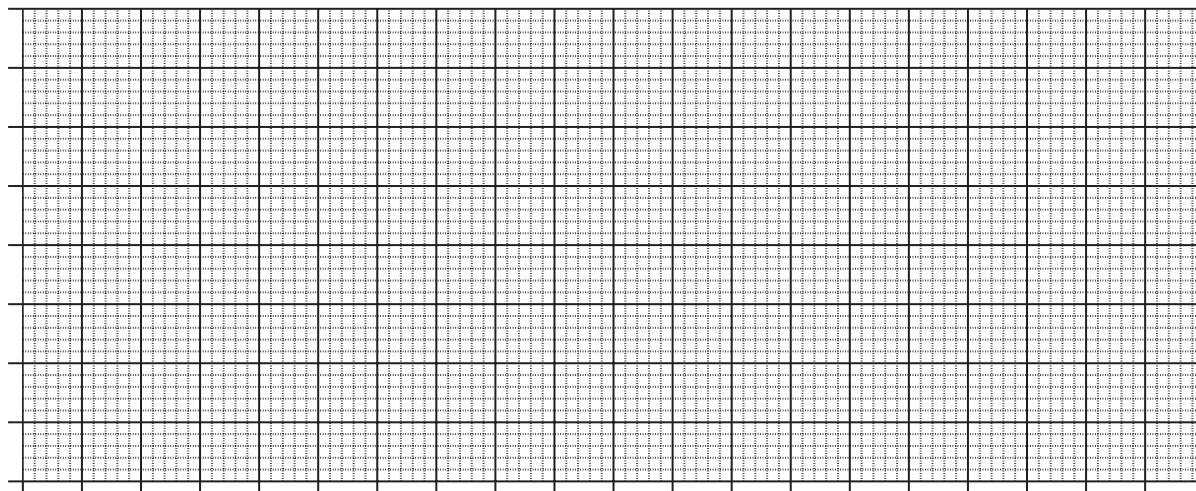
A student carried out a first-hand investigation to identify the relationship between the mass of a metal used in a reaction, and the volume of gas produced. The first-hand investigation was carried out at 25°C, and 100 kPa. In each experiment, 200 mL of hydrochloric acid was added to some zinc, and the volume of gas produced was recorded. The diagram shows the equipment used, and the table contains the student's results.

<i>Mass of zinc (g)</i>	<i>Volume of gas collected at 25°C and 100 kPa (mL)</i>
0.12	45
0.33	125
0.56	115
0.83	315
0.96	365
1.22	380
1.64	380
1.93	380



(a) Graph the results from these experiments.

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Question 22 continues on page 19

Question 22 (continued)

- (b) Predict the volume of gas that would be produced in this experiment if 3.00 g of zinc had been used. Justify your answer. 2

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- (c) Calculate the theoretical volume of gas produced at 25°C and 100 kPa by the reaction of 0.56 g of zinc with the 200 mL of hydrochloric acid. 2

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Question 23 (2 marks)

- (a) What causes incomplete combustion? 1

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- (b) Write a balanced equation for the incomplete combustion of methane. 1

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Question 24 (5 marks)

An antacid tablet is known to contain calcium carbonate (CaCO_3). To determine the mass of calcium carbonate in the tablet, the following procedure was used.

- The tablet was crushed and then placed in a beaker.
- A pipette was used to add 25.0 mL of 0.600 mol L^{-1} hydrochloric acid to the crushed tablet in the beaker.
- Once the reaction between the calcium carbonate and hydrochloric acid had stopped, phenolphthalein indicator was added to the reaction mixture.
- A teflon-coated burette was then used to add 0.100 mol L^{-1} sodium hydroxide to the beaker to neutralise the excess hydrochloric acid.
- The phenolphthalein changed from colourless to pink after 14.2 mL of the sodium hydroxide solution had been added.

- (a) Write a balanced chemical equation for the reaction that occurred between the calcium carbonate in the tablet and the hydrochloric acid. **1**

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- (b) How many moles of hydrochloric acid were added to the tablet? **1**

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- (c) Calculate the mass of calcium carbonate in the original antacid tablet. **3**

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Centre Number

Section I – Part B (continued)

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Student Number

Marks

Question 25 (6 marks)

A student collected a 500 mL sample of water from a local creek for analysis. It was filtered and the filtrate evaporated to dryness. The following data were collected.

Mass of filter paper	0.16 g
Mass of filter paper and solid	0.19 g
Mass of evaporating basin	45.33 g
Mass of basin and solid remaining	45.59 g

- (a) Calculate the percentage of total dissolved solids in the creek sample. 2

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- (b) It is suspected that lead has contaminated the creek. Describe a chemical test that could be carried out on the water sample to determine the presence of lead. 2

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- (c) The concentrations of ions in substances used by society need to be monitored. Justify this statement with reference to ONE ion you have studied. 2

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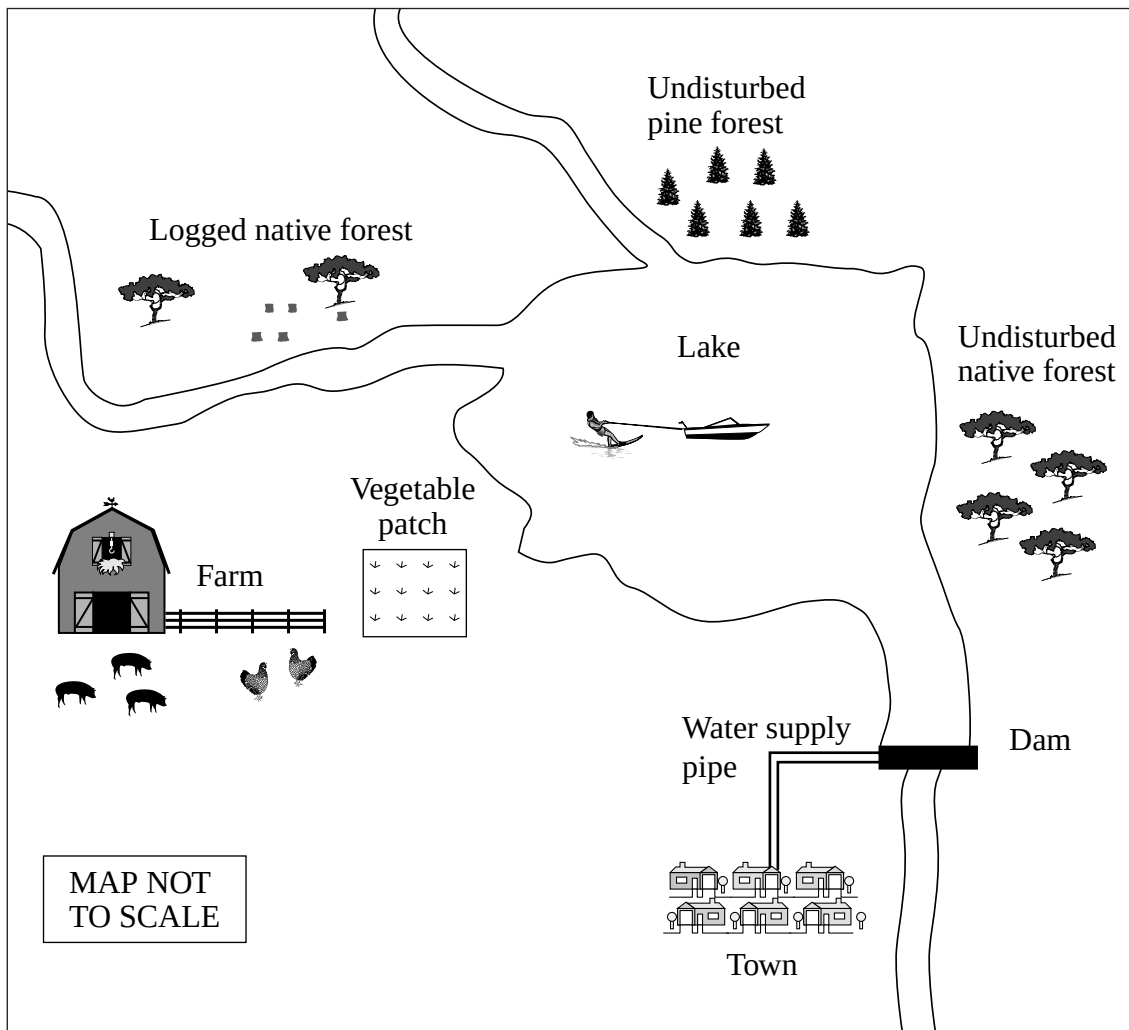
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Question 26 (7 marks)

The map shows the catchment for a town water supply.



Question 26 continues on page 23

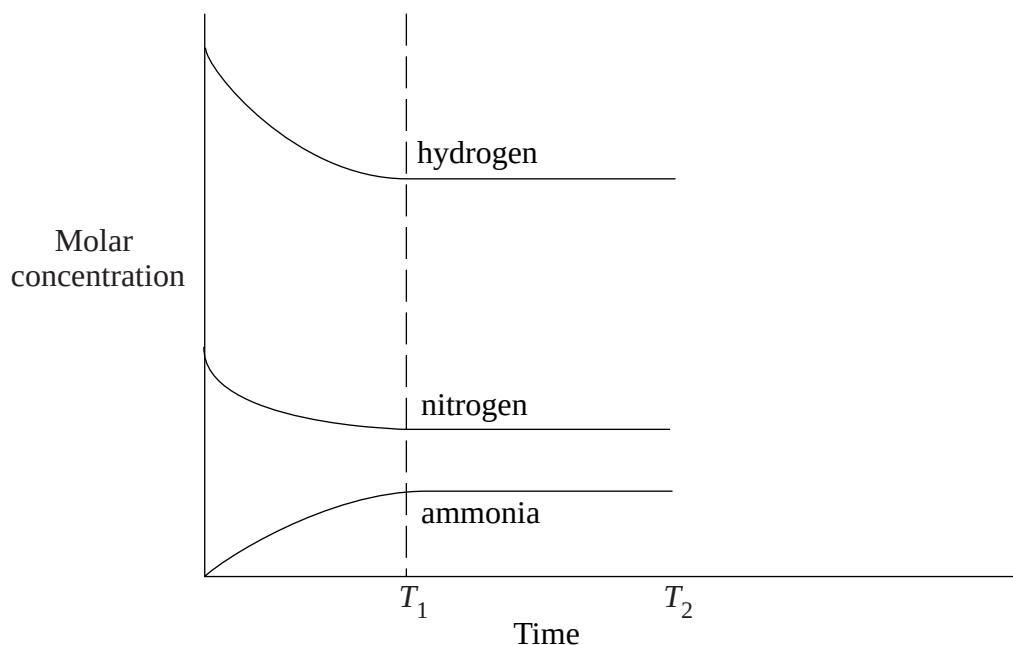
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– 23 –

Question 27 (5 marks)

The graph shows the variation in concentrations of reactants and product with time for the Haber process.



- (a) State why the concentrations of reactants and product do not change between T_1 and T_2 . **1**

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- (b) At time T_2 the volume of the reaction vessel was reduced.

- (i) Sketch on the graph how the concentrations of reactants and product would change after the volume was reduced. **2**

- (ii) Explain the changes shown on your graph. **2**

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Chemistry

Section II

25 marks

Attempt ONE question from Questions 28–32

Allow about 45 minutes for this section

Answer the question in a writing booklet. Extra writing booklets are available.

Show all relevant working in questions involving calculations.

	Pages
Question 28 Industrial Chemistry	26–27
Question 29 Shipwrecks, Corrosion and Conservation	28–29
Question 30 The Biochemistry of Movement	30–32
Question 31 The Chemistry of Art	33–34
Question 32 Forensic Chemistry	35–36

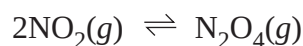
Question 28 — Industrial Chemistry (25 marks)

- (a) Sulfuric acid is an important industrial chemical. One of the intermediate products in the synthesis of sulfuric acid is $\text{H}_2\text{S}_2\text{O}_7$.

(i) What is the name of $\text{H}_2\text{S}_2\text{O}_7$? 1

(ii) With reference to the properties of sulfuric acid, describe safety precautions necessary for its transport. 3

- (b) Nitrogen dioxide forms an equilibrium mixture with dinitrogen tetraoxide as shown.



At 100°C , K for this reaction is 2.08.

At 25°C , a 1.00 L vessel initially contained 0.132 mol of $\text{NO}_2(g)$. Once equilibrium had been established, there was 0.0400 mol of $\text{N}_2\text{O}_4(g)$ in the vessel.

(i) Explain the effects of the addition of a catalyst and an increase in pressure on the yield of N_2O_4 in this reaction when carried out at 25°C . 2

(ii) Calculate the equilibrium constant for this reaction at 25°C , and account for any difference from the K value at 100°C . 4

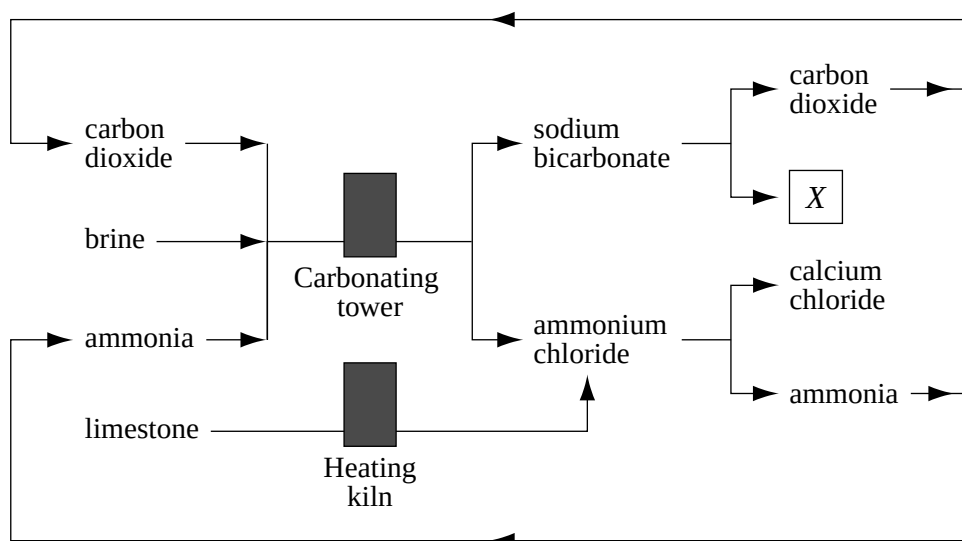
- (c) Advances in chemistry have impacted on the development of technologies. 7

Discuss this statement by analysing changes that have occurred in industrial methods for the production of sodium hydroxide.

Question 28 continues on page 27

Question 28 (continued)

(d) The diagram summarises the Solvay process.



(i) Identify compound X. 1

(ii) During the Solvay process, ammonia is recovered. 2

Outline the chemistry involved in the recovery of ammonia, and suggest a reason why this recovery is desirable.

(iii) Identify possible environmental issues that may arise from the Solvay process, and evaluate methods of dealing with those environmental issues. 5

End of Question 28

Question 29 — Shipwrecks, Corrosion and Conservation (25 marks)

- (a) (i) Reactive metals such as aluminium and zinc form a strongly adhering oxide coating as a result of the reaction with oxygen. **1**

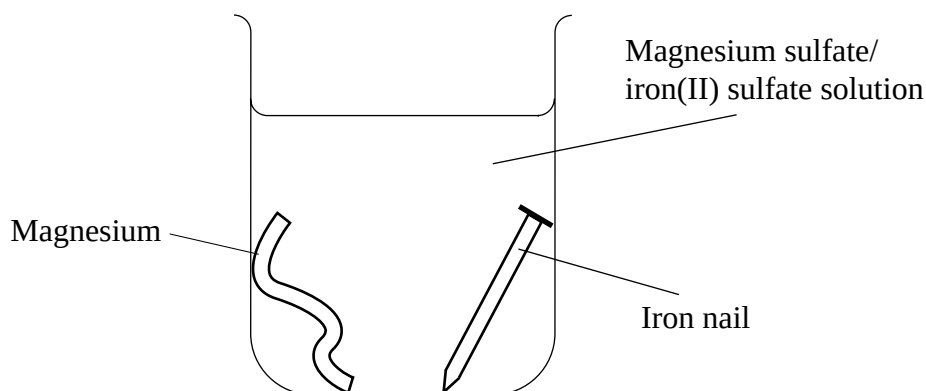
What name is given to these metals?

- (ii) The hulls of modern ships contain steels that have lower levels of carbon and sulfur than was used in the past. **3**

Compare the properties and uses of TWO types of steel that you have studied.

- (b) (i) Outline ONE factor that influences the rate of corrosion of iron in deep-water wrecks. **2**

- (ii) A student set up a laboratory experiment in which a beaker contained a strip of magnesium, an iron nail, and a solution of iron(II) sulfate and magnesium sulfate. This was left for several days. **4**



Describe what would happen in this experiment. Include relevant chemical equations in your answer.

Question 29 continues on page 29

Question 29 (continued)

- (c) Advances in chemistry have impacted on the development of technologies. 7

Discuss this statement by analysing current methods of corrosion protection for ships.

- (d) An iron cannon was recovered from a wreck on the Great Barrier Reef.



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The process used to restore the cannon was as follows:

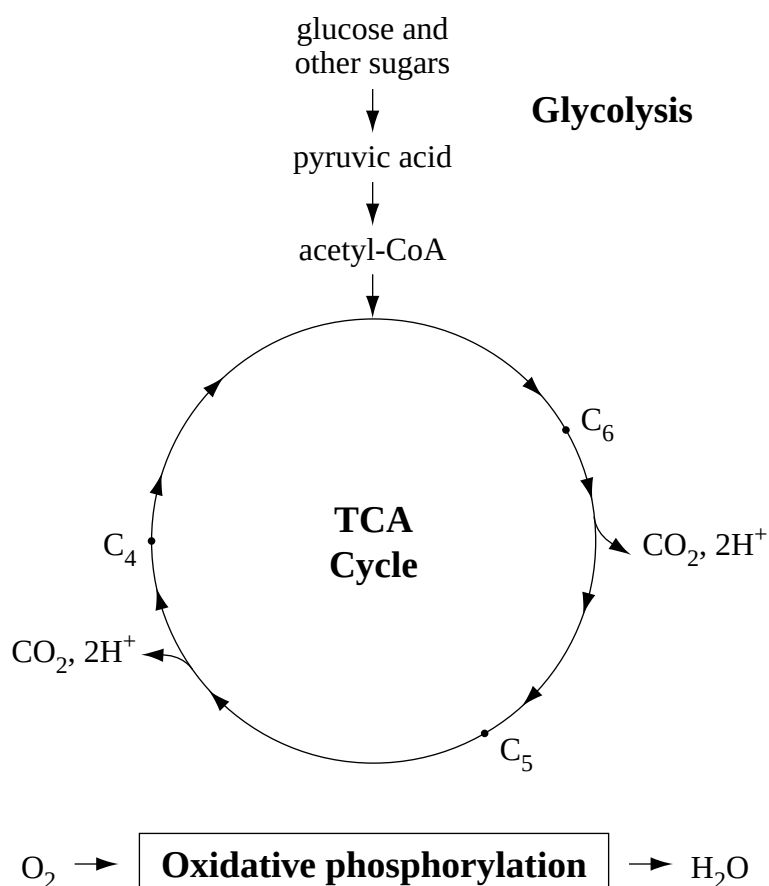
- Step 1: The cannon was placed in dilute sodium hydroxide for several weeks. During this period, the solution was replaced by fresh dilute sodium hydroxide solution at regular intervals.
- Step 2: Coral that had grown on the cannon was carefully chipped away.
- Step 3: Electrolysis was performed on the cannon.
- Step 4: The restored cannon was then coated in wax.

- (i) Why was the restored cannon coated in wax? 1
- (ii) Explain the purpose of Step 1 in the restoration of the iron cannon. 2
- (iii) Describe and justify the procedure for Step 3. Include relevant chemical equations in your answer. 5

End of Question 29

Question 30 — The Biochemistry of Movement (25 marks)

- (a) (i) Draw the structure of glycerol. 1
- (ii) Explain the viscosity of glycerol and its solubility in water. 3
- (b) The diagram below summarises some of the steps involved in respiration.

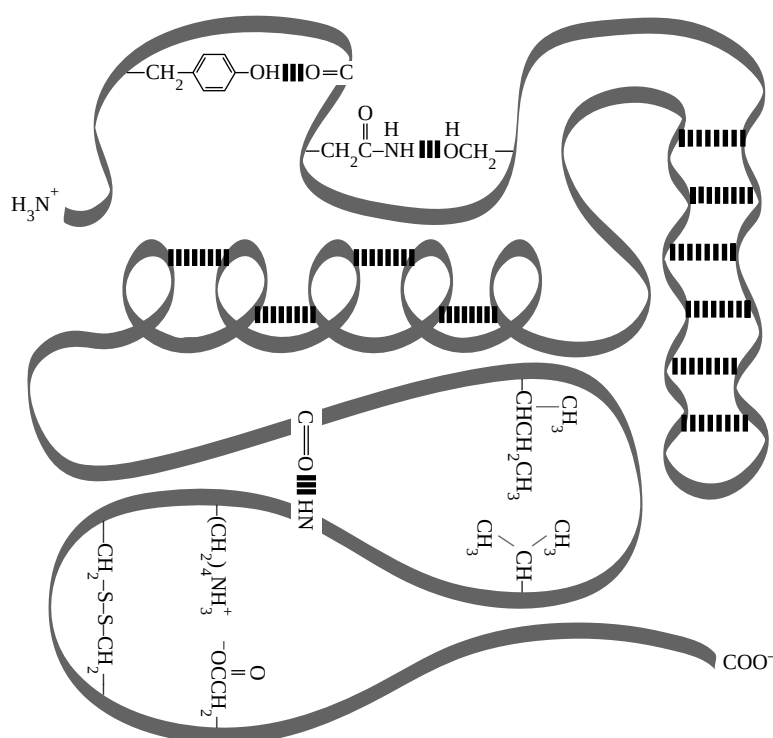


- (i) Compare the energy output from glycolysis with the total energy output from respiration. 2
- (ii) Explain the role of oxidation and reduction in respiration. 4

Question 30 continues on page 31

Question 30 (continued)

(c) A diagram illustrating the structure of a protein is shown.



Describe how chemical features affect the structure and shape of proteins, and assess the importance of shape in the current model of enzyme function.

7

Question 30 continues on page 32

Question 30 (continued)

- (d) An athletics coach persuaded his runners to drink one of two sports drinks immediately before a race to boost performance. The composition of the two drinks is shown in the table. The coach believes that the sports drinks will give the runners extra energy during both a sprint race and a marathon.

<i>Name of sports drink</i>	<i>Carbohydrate content (%)</i>	<i>Carbohydrate composition</i>	<i>Sodium (mg per 500 mL)</i>
Athleaid	6	sucrose glucose fructose	225
Energade	8	fructose maltodextrin*	130

*Maltodextrin is a polymer composed of many glucose molecules bonded together.

- (i) Define the term *carbohydrate*. 1
- (ii) Describe the chemical process involved when humans store carbohydrates for later use. 2
- (iii) Use data in the table to assess the hypothesis made by the coach. 5

End of Question 30

Question 31 — The Chemistry of Art (25 marks)

(a) A solution of iron(II) sulfate is mixed with a solution of potassium permanganate under acidic conditions, resulting in a colour change.

(i) What is the observed colour change? **1**

(ii) Account for the colour change in terms of oxidation states. Include relevant half-equations in your answer. **3**

(b) The table provides examples of the colours shown by transition metal complexes.

<i>Metal ion</i>	<i>Ligand</i>	<i>Colour of complex</i>
Cu^{2+}	H_2O	light blue
Cu^{2+}	NH_3	dark blue
Cr^{3+}	H_2O	green
Cr^{2+}	H_2O	violet

(i) Account for the differences in colours shown by these transition metal complexes. **2**

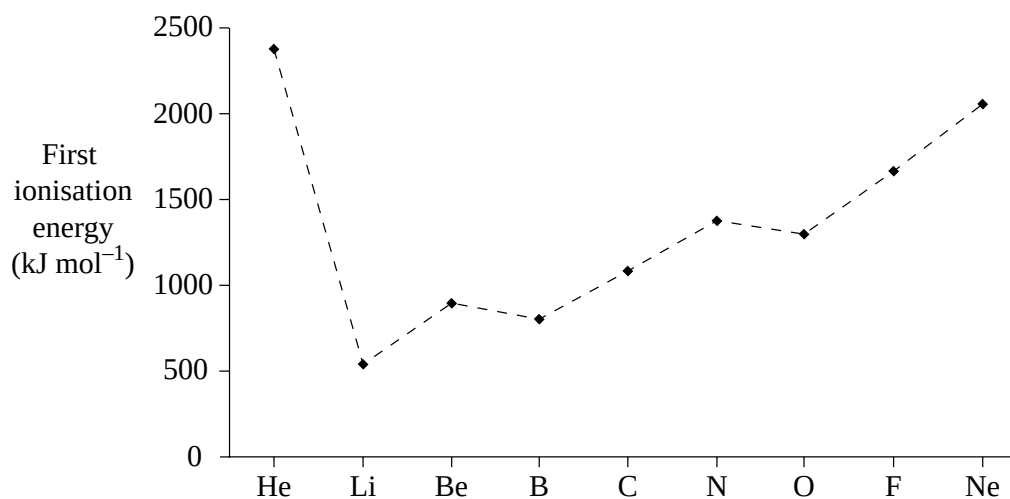
(ii) Assess the importance of minerals as pigments to TWO ancient cultures. **4**

(c) Describe the methodology and use of laser microspectral analysis, and assess the impact of advances in the understanding of emission spectra on the development of technologies. **7**

Question 31 continues on page 34

Question 31 (continued)

- (d) (i) Define the Pauli exclusion principle. **1**
- (ii) Distinguish between first ionisation energy and electronegativity. **2**
- (iii) The graph shows the first ionisation energies for the elements helium to neon. **5**



Analyse the relationship between first ionisation energy and electron configuration, using data from the graph.

End of Question 31

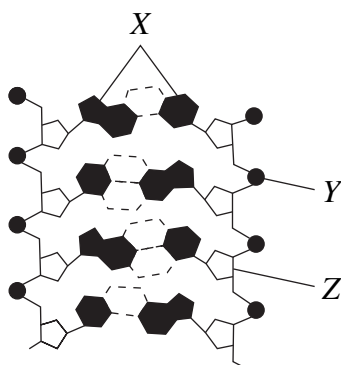
Question 32 — Forensic Chemistry (25 marks)

- (a) The diagram shows the emission spectra of three elements and an unknown mixture.



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- (i) Name TWO elements present in the unknown mixture. 1
- (ii) Describe how emission spectra assist in the identification of elements in a mixture. 3
- (b) The diagram illustrates the structure of DNA.



- (i) Identify the components labelled X, Y and Z. 1
- (ii) Explain how the analysis of DNA allows for identification of relationships between people. 2
- (iii) Discuss the ethics of maintaining data banks of DNA. 3

Question 32 continues on page 36

Question 32 (continued)

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|-----|--|----------|
| (c) | Describe the structure and composition of proteins, and assess the potential of electrophoresis as a method for their analysis. | 7 |
| (d) | (i) What type of carbohydrate is sucrose? | 1 |
| | (ii) Describe a chemical test that can be used to distinguish glucose from starch. | 2 |
| | (iii) A scientist wanted to develop a method of separation for a mixture of six carbohydrates using high performance liquid chromatography. The diagram shows the chromatograms obtained for the mixture using two different solvents as the mobile phase. | 5 |

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Describe high performance liquid chromatography, and assess the effectiveness of the different solvents used by the scientist for separating the mixture of carbohydrates.

End of paper

DATA SHEET

Avogadro constant, N_A	$6.022 \times 10^{23} \text{ mol}^{-1}$
Volume of 1 mole ideal gas: at 100 kPa and	
at 0°C (273.15 K)	22.71 L
at 25°C (298.15 K)	24.79 L
Ionisation constant for water at 25°C (298.15 K), K_w	1.0×10^{-14}
Specific heat capacity of water	$4.18 \times 10^3 \text{ J kg}^{-1} \text{ K}^{-1}$

Some useful formulae

$$\text{pH} = -\log_{10}[\text{H}^+]$$

$$\Delta H = -m C \Delta T$$

Some standard potentials

$\text{K}^+ + \text{e}^-$	$\rightleftharpoons$	K(s)	-2.94 V
$\text{Ba}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Ba(s)	-2.91 V
$\text{Ca}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Ca(s)	-2.87 V
$\text{Na}^+ + \text{e}^-$	$\rightleftharpoons$	Na(s)	-2.71 V
$\text{Mg}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Mg(s)	-2.36 V
$\text{Al}^{3+} + 3\text{e}^-$	$\rightleftharpoons$	Al(s)	-1.68 V
$\text{Mn}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Mn(s)	-1.18 V
$\text{H}_2\text{O} + \text{e}^-$	$\rightleftharpoons$	$\frac{1}{2}\text{H}_2(\text{g}) + \text{OH}^-$	-0.83 V
$\text{Zn}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Zn(s)	-0.76 V
$\text{Fe}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Fe(s)	-0.44 V
$\text{Ni}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Ni(s)	-0.24 V
$\text{Sn}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Sn(s)	-0.14 V
$\text{Pb}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Pb(s)	-0.13 V
$\text{H}^+ + \text{e}^-$	$\rightleftharpoons$	$\frac{1}{2}\text{H}_2(\text{g})$	0.00 V
$\text{SO}_4^{2-} + 4\text{H}^+ + 2\text{e}^-$	$\rightleftharpoons$	$\text{SO}_2(\text{aq}) + 2\text{H}_2\text{O}$	0.16 V
$\text{Cu}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Cu(s)	0.34 V
$\frac{1}{2}\text{O}_2(\text{g}) + \text{H}_2\text{O} + 2\text{e}^-$	$\rightleftharpoons$	2OH^-	0.40 V
$\text{Cu}^+ + \text{e}^-$	$\rightleftharpoons$	Cu(s)	0.52 V
$\frac{1}{2}\text{I}_2(\text{s}) + \text{e}^-$	$\rightleftharpoons$	I^-	0.54 V
$\frac{1}{2}\text{I}_2(\text{aq}) + \text{e}^-$	$\rightleftharpoons$	I^-	0.62 V
$\text{Fe}^{3+} + \text{e}^-$	$\rightleftharpoons$	Fe^{2+}	0.77 V
$\text{Ag}^+ + \text{e}^-$	$\rightleftharpoons$	Ag(s)	0.80 V
$\frac{1}{2}\text{Br}_2(\text{l}) + \text{e}^-$	$\rightleftharpoons$	Br^-	1.08 V
$\frac{1}{2}\text{Br}_2(\text{aq}) + \text{e}^-$	$\rightleftharpoons$	Br^-	1.10 V
$\frac{1}{2}\text{O}_2(\text{g}) + 2\text{H}^+ + 2\text{e}^-$	$\rightleftharpoons$	H_2O	1.23 V
$\frac{1}{2}\text{Cl}_2(\text{g}) + \text{e}^-$	$\rightleftharpoons$	Cl^-	1.36 V
$\frac{1}{2}\text{Cr}_2\text{O}_7^{2-} + 7\text{H}^+ + 3\text{e}^-$	$\rightleftharpoons$	$\text{Cr}^{3+} + \frac{7}{2}\text{H}_2\text{O}$	1.36 V
$\frac{1}{2}\text{Cl}_2(\text{aq}) + \text{e}^-$	$\rightleftharpoons$	Cl^-	1.40 V
$\text{MnO}_4^- + 8\text{H}^+ + 5\text{e}^-$	$\rightleftharpoons$	$\text{Mn}^{2+} + 4\text{H}_2\text{O}$	1.51 V
$\frac{1}{2}\text{F}_2(\text{g}) + \text{e}^-$	$\rightleftharpoons$	F^-	2.89 V

PERIODIC TABLE OF THE ELEMENTS

PERIODIC TABLE OF THE ELEMENTS

1 H 1.008 Hydrogen		KEY										2 He 4.003 Helium			
3 Li 6.941 Lithium		4 Be 9.012 Beryllium		Atomic Number		Atomic Weight		79 Au 197.0 Gold		Symbol of element		8 O 16.00 Oxygen		9 F 19.00 Fluorine	
		12 Mg 24.31 Magnesium		25 Mn 54.94 Manganese		26 Fe 55.85 Iron		27 Co 58.93 Cobalt		28 Ni 58.69 Nickel		30 Zn 65.41 Zinc		35 Br 79.90 Bromine	
11 Na 22.99 Sodium		23 V 50.94 Vanadium		24 Cr 52.00 Chromium		25 Mn 54.94 Manganese		26 Fe 55.85 Iron		27 Co 58.93 Cobalt		28 Ni 58.69 Nickel		30 Zn 65.41 Zinc	
19 K 39.10 Potassium		21 Sc 44.96 Scandium		22 Ti 47.87 Titanium		23 V 50.94 Vanadium		24 Cr 52.00 Chromium		25 Mn 54.94 Manganese		26 Fe 55.85 Iron		27 Co 58.93 Cobalt	
37 Rb 85.47 Rubidium		39 Y 88.91 Yttrium		40 Zr 91.22 Zirconium		41 Nb 92.91 Niobium		42 Mo 95.94 Molybdenum		43 Tc [97.91] Technetium		44 Ru 101.1 Ruthenium		45 Rh 102.9 Rhodium	
55 Cs 132.9 Caesium		57–71 Lanthanides		72 Hf 178.5 Hafnium		73 Ta 180.9 Tantalum		74 W 183.8 Tungsten		75 Re 186.2 Rhenium		76 Os 190.2 Osmium		77 Ir 192.2 Iridium	
87 Fr [223.0] Francium		89–103 Actinides		104 Rf [261.1] Rutherfordium		105 Db [262.1] Dubnium		106 Sg [266.1] Seaborgium		107 Bh [264.1] Bohrium		108 Hs [277] Hassium		109 Mt [268] Meitnerium	
		88 Ra [226.0] Radium		104 Rf [261.1] Rutherfordium		105 Db [262.1] Dubnium		106 Sg [266.1] Seaborgium		107 Bh [264.1] Bohrium		108 Hs [277] Hassium		109 Mt [268] Meitnerium	
				110 Ds [271] Darmstadtium		111 Rg [272] Roentgenium		112 Cn [285] Copernicium		113 Nh [286] Nihonium		114 Fl [289] Flerovium		115 Mc [290] Moscovium	
				116 Lv [293] Livermorium		117 Ts [294] Tennessine		118 Og [294] Oganesson		119 Uue [295] Ununennium		120 Uub [296] Unbinilium		121 Uut [297] Untrium	
				122 Uuq [298] Unquadium		123 Uubk [299] Unbikium		124 Uuqk [300] Unbiquadium		125 Uutk [301] Unbikium		126 Uuqk [302] Unbiquadium		127 Uutk [303] Unbikium	
				128 Uuek [310] Unquadium		129 Uubk [311] Unbikium		130 Uuqk [312] Unbiquadium		131 Uutk [313] Unbikium		132 Uuqk [314] Unbiquadium		133 Uutk [315] Unbikium	
				134 Uuek [316] Unquadium		135 Uubk [317] Unbikium		136 Uuqk [318] Unbiquadium		137 Uutk [319] Unbikium		138 Uuqk [320] Unbiquadium		139 Uutk [321] Unbikium	
				140 Uuek [322] Unquadium		141 Uubk [323] Unbikium		142 Uuqk [324] Unbiquadium		143 Uutk [325] Unbikium		144 Uuqk [326] Unbiquadium		145 Uutk [327] Unbikium	
				146 Uuek [328] Unquadium		147 Uubk [329] Unbikium		148 Uuqk [330] Unbiquadium		149 Uutk [331] Unbikium		150 Uuqk [332] Unbiquadium		151 Uutk [333] Unbikium	
				152 Uuek [334] Unquadium		153 Uubk [335] Unbikium		154 Uuqk [336] Unbiquadium		155 Uutk [337] Unbikium		156 Uuqk [338] Unbiquadium		157 Uutk [339] Unbikium	
				158 Uuek [340] Unquadium		159 Uubk [341] Unbikium		160 Uuqk [342] Unbiquadium		161 Uutk [343] Unbikium		162 Uuqk [344] Unbiquadium		163 Uutk [345] Unbikium	
				164 Uuek [346] Unquadium		165 Uubk [347] Unbikium		166 Uuqk [348] Unbiquadium		167 Uutk [349] Unbikium		168 Uuqk [350] Unbiquadium		169 Uutk [351] Unbikium	
				170 Uuek [352] Unquadium		171 Uubk [353] Unbikium		172 Uuqk [354] Unbiquadium		173 Uutk [355] Unbikium		174 Uuqk [356] Unbiquadium		175 Uutk [357] Unbikium	
				176 Uuek [358] Unquadium		177 Uubk [359] Unbikium		178 Uuqk [360] Unbiquadium		179 Uutk [361] Unbikium		180 Uuqk [362] Unbiquadium		181 Uutk [363] Unbikium	
				182 Uuek [364] Unquadium		183 Uubk [365] Unbikium		184 Uuqk [366] Unbiquadium		185 Uutk [367] Unbikium		186 Uuqk [368] Unbiquadium		187 Uutk [369] Unbikium	
				188 Uuek [370] Unquadium		189 Uubk [371] Unbikium		190 Uuqk [372] Unbiquadium		191 Uutk [373] Unbikium		192 Uuqk [374] Unbiquadium		193 Uutk [375] Unbikium	
				194 Uuek [376] Unquadium		195 Uubk [377] Unbikium		196 Uuqk [378] Unbiquadium		197 Uutk [379] Unbikium		198 Uuqk [380] Unbiquadium		199 Uutk [381] Unbikium	
				200 Uuek [382] Unquadium		201 Uubk [383] Unbikium		202 Uuqk [384] Unbiquadium		203 Uutk [385] Unbikium		204 Uuqk [386] Unbiquadium		205 Uutk [387] Unbikium	
				206 Uuek [388] Unquadium		207 Uubk [389] Unbikium		208 Uuqk [390] Unbiquadium		209 Uutk [391] Unbikium		210 Uuqk [392] Unbiquadium		211 Uutk [393] Unbikium	
				212 Uuek [394] Unquadium		213 Uubk [395] Unbikium		214 Uuqk [396] Unbiquadium		215 Uutk [397] Unbikium		216 Uuqk [398] Unbiquadium		217 Uutk [399] Unbikium	
				218 Uuek [399] Unquadium		219 Uubk [400] Unbikium		220 Uuqk [401] Unbiquadium		221 Uutk [402] Unbikium		222 Uuqk [403] Unbiquadium		223 Uutk [404] Unbikium	
				224 Uuek [405] Unquadium		225 Uubk [406] Unbikium		226 Uuqk [407] Unbiquadium		227 Uutk [408] Unbikium		228 Uuqk [409] Unbiquadium		229 Uutk [410] Unbikium	
				230 Uuek [410] Unquadium		231 Uubk [411] Unbikium		232 Uuqk [412] Unbiquadium		233 Uutk [413] Unbikium		234 Uuqk [414] Unbiquadium		235 Uutk [415] Unbikium	
				236 Uuek [416] Unquadium		237 Uubk [417] Unbikium		238 Uuqk [418] Unbiquadium		239 Uutk [419] Unbikium		240 Uuqk [420] Unbiquadium		241 Uutk [421] Unbikium	
				242 Uuek [420] Unquadium		243 Uubk [421] Unbikium		244 Uuqk [422] Unbiquadium		245 Uutk [423] Unbikium		246 Uuqk [424] Unbiquadium		247 Uutk [425] Unbikium	
				248 Uuek [427] Unquadium		249 Uubk [428] Unbikium		250 Uuqk [429] Unbiquadium		251 Uutk [430] Unbikium		252 Uuqk [431] Unbiquadium		253 Uutk [432] Unbikium	
				254 Uuek [434] Unquadium		255 Uubk [435] Unbikium		256 Uuqk [436] Unbiquadium		257 Uutk [437] Unbikium		258 Uuqk [438] Unbiquadium		259 Uutk [439] Unbikium	
				260 Uuek [440] Unquadium		261 Uubk [441] Unbikium		262 Uuqk [442] Unbiquadium		263 Uutk [443] Unbikium		264 Uuqk [444] Unbiquadium		265 Uutk [445] Unbikium	
				266 Uuek [448] Unquadium		267 Uubk [449] Unbikium		268 Uuqk [450] Unbiquadium		269 Uutk [451] Unbikium		270 Uuqk [452] Unbiquadium		271 Uutk [453] Unbikium	
				272 Uuek [454] Unquadium		273 Uubk [455] Unbikium		274 Uuqk [456] Unbiquadium		275 Uutk [457] Unbikium		276 Uuqk [458] Unbiquadium		277 Uutk [459] Unbikium	
				278 Uuek [460] Unquadium		279 Uubk [461] Unbikium		280 Uuqk [462] Unbiquadium		281 Uutk [463] Unbikium		282 Uuqk [464] Unbiquadium		283 Uutk [465] Unbikium	
				284 Uuek [468] Unquadium		285 Uubk [469] Unbikium		286 Uuqk [470] Unbiquadium		287 Uutk [471] Unbikium		288 Uuqk [472] Unbiquadium		289 Uutk [473] Unbikium	
				290 Uuek [474] Unquadium		291 Uubk [475] Unbikium		292 Uuqk [476] Unbiquadium		293 Uutk [477] Unbikium		294 Uuqk [478] Unbiquadium		295 Uutk [479] Unbikium	
				296 Uuek [480] Unquadium		297 Uubk [481] Unbikium		298 Uuqk [482] Unbiquadium		299 Uutk [483] Unbikium		300 Uuqk [484] Unbiquadium		301 Uutk [485] Unbikium	
				302 Uuek [486] Unquadium		303 Uubk [487] Unbikium		304 Uuqk [488] Unbiquadium		305 Uutk [489] Unbikium		306 Uuqk [490] Unbiquadium		307 Uutk [491] Unbikium	
				308 Uuek [490] Unquadium		309 Uubk [491] Unbikium		310 Uuqk [492] Unbiquadium		311 Uutk [493] Unbikium		312 Uuqk [494] Unbiquadium		313 Uutk [495] Unbikium	
				314 Uuek [494] Unquadium		315 Uubk [495] Unbikium		316 Uuqk [496] Unbiquadium		317 Uutk [497] Unbikium		318 Uuqk [498] Unbiquadium		319 Uutk [499] Unbikium	
				320 Uuek [498] Unquadium		321 Uubk [499] Unbikium		322 Uuqk [500] Unbiquadium		323 Uutk [501] Unbikium		324 Uuqk [502] Unbiquadium		325 Uutk [503] Unbikium	
				326 Uuek [502] Unquadium		327 Uubk [503] Unbikium		328 Uuqk [504] Unbiquadium		329 Uutk [505] Unbikium		330 Uuqk [506] Unbiquadium		331 Uutk [507] Unbikium	
				332 Uuek [506] Unquadium		333 Uubk [507] Unbikium		334 Uuqk [508] Unbiquadium		335 Uutk [509] Unbikium		336 Uuqk [510] Unbiquadium		337 Uutk [511] Unbikium	
				338 Uuek [510] Unquadium		339 Uubk [511] Unbikium		340 Uuqk [512] Unbiquadium		341 Uutk [513] Unbikium		342 Uuqk [514] Unbiquadium		343 Uutk [515] Unbikium	
				344 Uuek [514] Unquadium		345 Uubk [515] Unbikium		346 Uuqk [516] Unbiquadium		347 Uutk [517] Unbikium		348 Uuqk [518] Unbiquadium		349 Uutk [519] Unbikium	
				350 Uuek [518] Unquadium		351 Uubk [519] Unbikium		352 Uuqk [520] Unbiquadium		353 Uutk [521] Unbikium		354 Uuqk [522] Unbiquadium		355 Uutk [523] Unbikium	
				356 Uuek [522] Unquadium		357 Uubk [523] Unbikium		358 Uuqk [524] Unbiquadium		359 Uutk [525] Unbikium		360 Uuqk [526] Unbiquadium		361 Uutk [527] Unbikium	
				362 Uuek [526] Unquadium		363 Uubk [527] Unbikium		364 Uuqk [528] Unbiquadium		365 Uutk [529] Unbikium		366 Uuqk [530] Unbiquadium		367 Uutk [531] Unbikium	
				368 Uuek [530] Unquadium		369 Uubk [531] Unbikium		370 Uuqk [532] Unbiquadium		371 Uutk [533] Unbikium		372 Uuqk [534] Unbiquadium		373 Uutk [535] Unbikium	
				374 Uuek [534] Unquadium		375 Uubk [535] Unbikium		376 Uuqk [536] Unbiquadium		377 Uutk [537] Unbikium		378 Uuqk [538] Unbiquadium		379 Uutk [539] Unbikium	
				380 Uuek [538] Unquadium		381 Uubk [539] Unbikium		382 Uuqk [540] Unbiquadium		383 Uutk [541] Unbikium		384 Uuqk [542] Unbiquadium		385 Uutk [543] Unbikium	
				386 Uuek [542] Unquadium		387 Uubk [543] Unbikium		388 Uuqk [544] Unbiquadium		389 Uutk [545] Unbikium		390 Uuqk [546] Unbiquadium		391 Uutk [547] Unbikium	
				392 Uuek [546] Unquadium		393 Uubk [547] Unbikium		394 Uuqk [548] Unbiquadium		395 Uutk [549] Unbikium		396 Uuqk [550] Unbiquadium		397 Uutk [551] Unbikium	
				398 Uuek [550] Unquadium		399 Uubk [551] Unbikium		400 Uuqk [552] Unbiquadium		401 Uutk [553] Unbikium		402 Uuqk [554] Unbiquadium		403 Uutk [555] Unbikium	
				404 Uuek [554] Unquadium		405 Uubk [555] Unbikium		406 Uuqk [556] Unbiquadium		407 Uutk [557] Unbikium		408 Uuqk [558] Unbiquadium		409 Uutk [559] Unbikium	
				410 Uuek [558] Unquadium		411 Uubk [559] Unbikium		412 Uuqk [560] Unbiquadium		413 Uutk [561] Unbikium		414 Uuqk [562] Unbiquadium		415 Uutk [563] Unbikium	
				416 Uuek [562] Unquadium		417 Uubk [563] Unbikium		418 Uuqk [564] Unbiquadium		419 Uutk [565] Unbikium		420 Uuqk [566] Unbiquadium		421 Uutk [567] Unbikium	
				422 Uuek [566] Unquadium		423 Uubk [567] Unbikium		424 Uuqk [568] Unbiquadium		425 Uutk [569] Unbikium		426 Uuqk [570] Unbiquadium		427 Uutk [571] Unbikium	
				428 Uuek [570] Unquadium		429 Uubk [571] Unbikium		430 Uuqk [572] Unbiquadium		431 Uutk [573] Unbikium		432 Uuqk [574] Unbiquadium		433 Uutk [575] Unbikium	
				434 Uuek [574] Unquadium		435 Uubk [575] Unbikium		436 Uuqk [576] Unbiquadium		437 Uutk [577] Unbikium		438 Uuqk [578] Unbiquadium		439 Uutk [579] Unbikium	
				440 Uuek [578] Unquadium		441 Uubk [579] Unbikium		442 Uuqk [580] Unbiquadium		443 Uutk [581] Unbikium		444 Uuqk [582]<			

Lanthanides

57 La 138.9	58 Ce 140.1	59 Pr 140.9	60 Nd 144.2	61 Pm [144.9]	62 Sm 150.4	63 Eu 152.0	64 Gd 157.3	65 Tb 158.9	66 Dy 162.5	67 Ho 164.9	68 Er 167.3	69 Tm 168.9	70 Yb 173.0	71 Lu 175.0
Lanthanum	Cerium	Praseodymium	Neodymium	Promethium	Samarium	Europium	Gadolinium	Terbium	Dysprosium	Holmium	Erbium	Thulium	Ytterbium	Lutetium

Actinides

89	90	91	92	93	94	95	96	97	98	99	100	101	102	103
Ac	Th	Pa	U	Np	Pu	Am	Cm	Bk	Cf	Es	Fm	Md	No	Lr
Actinium	Thorium	Protactinium	Uranium	Neptunium	Plutonium	Americium	Curium	Berkelium	Californium	Einsteinium	Fermium	Mendelevium	Nobelium	Lawrencium
[227.0]	232.0	231.0	238.0	[237.0]	[244.1]	[243.1]	[247.1]	[247.1]	[251.1]	[252.1]	[257.1]	[258.1]	[259.1]	[262.1]

Where the atomic weight is not known, the relative atomic mass of the most common radioactive isotope is shown in brackets.

The atomic weights of Np and Tc are given for the isotopes ^{237}Np and ^{99}Tc .